
Kubernetes Health Probes – Explained

Kubernetes uses **probes** to check the health of containers running inside pods. This helps Kubernetes decide **when to restart a container**, **when to remove it from service**, and **when it's ready to receive traffic**.

Types of Health Probes

There are **three types** of health probes in Kubernetes:

Probe Type	Purpose
Liveness Probe	Checks if the app is alive (running). If it fails, the container is restarted .
Readiness Probe	Checks if the app is ready to serve requests. If it fails, the pod is removed from the Service load balancer .
Startup Probe	Used to give slow-starting apps more time. Prevents liveness probe from killing the app too soon.

How Probes Work

Probes can use one of the following mechanisms:

Mechanism	Description
HTTP GET	Calls a specified HTTP endpoint (e.g., <code>/healthz</code>) inside the container.
TCP Socket	Tries to establish a TCP connection on a specific port.
Exec Command	Runs a command in the container. If it returns exit code <code>0</code> , it's healthy.

Liveness Probe Example (HTTP)

livenessProbe:

httpGet:

path: /healthz

port: 8080

initialDelaySeconds: 3

periodSeconds: 10

- **initialDelaySeconds:** Wait 3 seconds before first probe.
 - **periodSeconds:** Check every 10 seconds.
 - If `/healthz` fails, the container is **restarted**.
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Readiness Probe Example (TCP)

readinessProbe:

tcpSocket:

port: 3306

initialDelaySeconds: 5

periodSeconds: 10

- Checks if the database port is open.
 - If it fails, **pod is removed from Service endpoints**.
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Startup Probe Example (Exec)

startupProbe:

exec:

command:

- cat

- /tmp/healthy

initialDelaySeconds: 5
periodSeconds: 10
failureThreshold: 30

- Gives the app up to $30 \times 10 = 300$ seconds (5 minutes) to start.
 - Useful when containers take time to become ready (e.g., Java apps).
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Best Practices

- Use **Startup Probe** for slow apps to avoid false liveness failures.
 - Liveness Probes should be **lightweight and fast**.
 - Readiness Probes should be accurate in indicating when the app is **ready for traffic**.
 - Always test your probe endpoints locally before applying them.
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What Happens When a Probe Fails?

Probe	What Happens
Liveness Probe	Container is restarted
Readiness Probe	Pod is removed from load balancing
Startup Probe	Until it succeeds, liveness probe is ignored

Probes in Action

If your app's `/health` endpoint returns **500**:

- Liveness Probe will restart the container (if configured).
 - Readiness Probe will mark it as **Not Ready** (removed from service).
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Real-Life Use Case

Java Spring Boot app

- Takes 60 seconds to initialize.
 - Use a **startupProbe** with high **failureThreshold**.
 - Use **readinessProbe** to check **/actuator/health**.
 - Use **livenessProbe** to catch hangs or memory leaks.
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